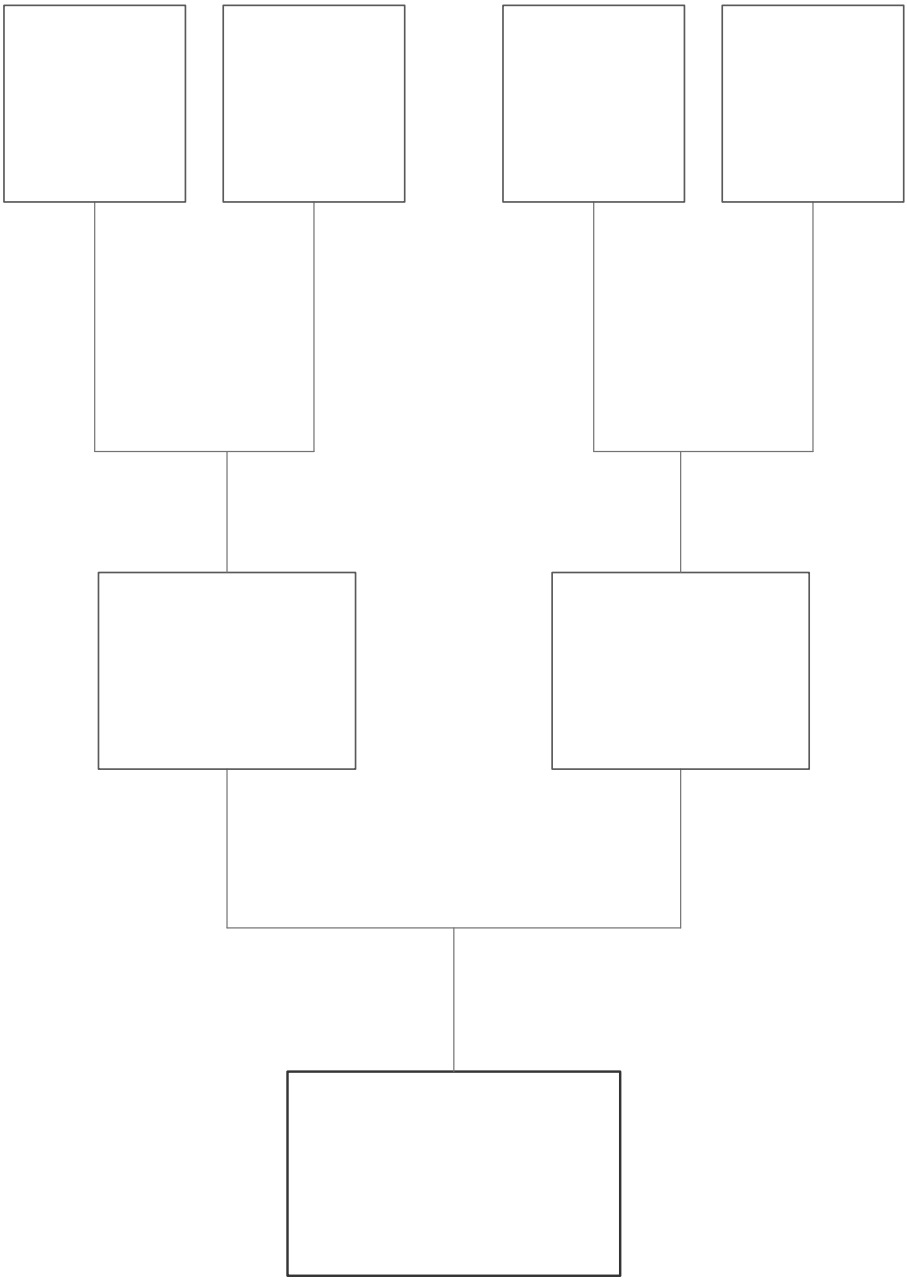Bottom-up tree